

2-Phase Interleaved Sync-Buck Converter

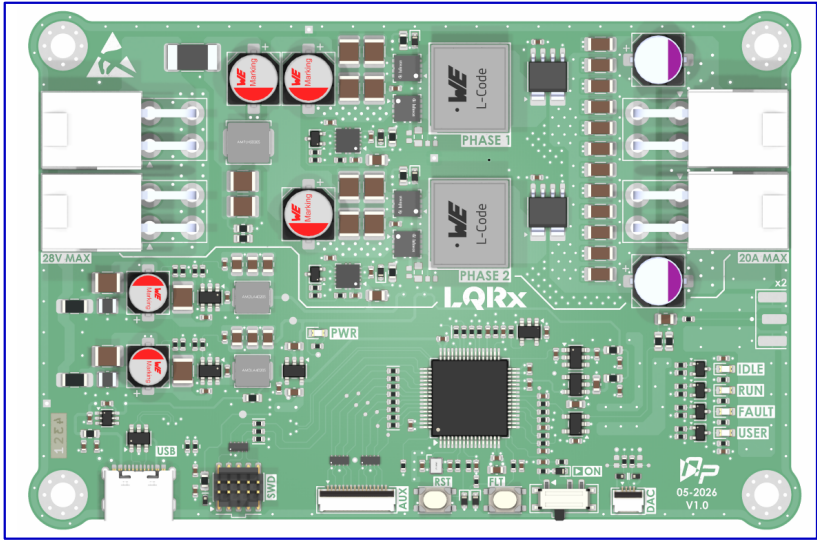
Digitally Controlled Power Conversion with STM32G474

Revision 1.0

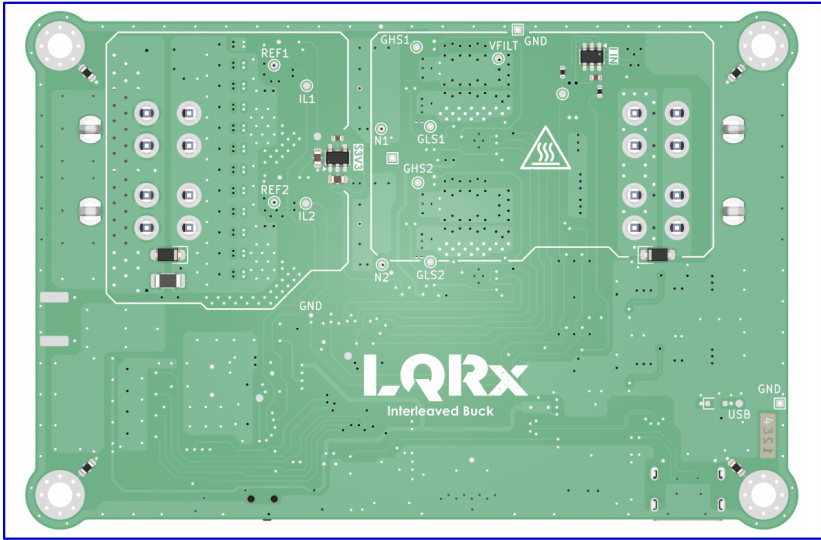
Table of Contents

- A Project Overview
- B Block Diagram
- C Top Hierarchical View
 - 1 Power State
 - 2 Power Connectors
 - 3 Power Supplies
 - 4 Microcontroller Power
 - 5 Microcontroller IOs
 - 6 Analog Front-End
 - 7 Interface Interconnects
 - 8 User LEDs
 - 9 User Interface
- D Power Supply Sequencing
- E Revision History

TOP VIEW



BOTTOM VIEW



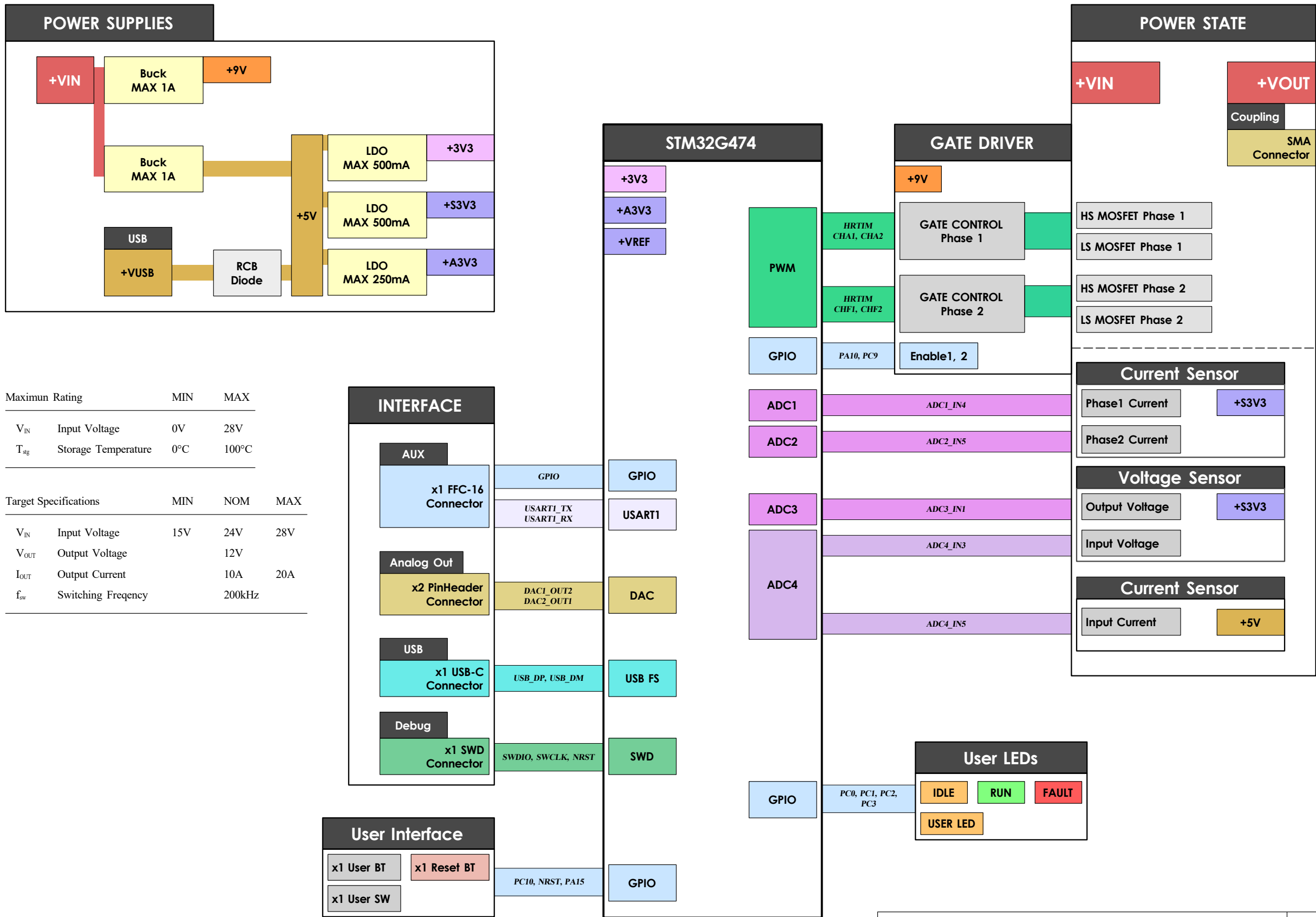
Legend

DESIGN NOTE: Text for information design.	DEBUG NOTE: Text for Debug notes.	DESIGN NOTE: Text for design guideline	DESIGN NOTE: Text for critical design	LAYOUT NOTE: Text for layout guideline	LAYOUT NOTE: Text for Critical layout	General Note Warning Note Text to be added to silkscreen.
--	--------------------------------------	---	--	---	--	---

LQRx

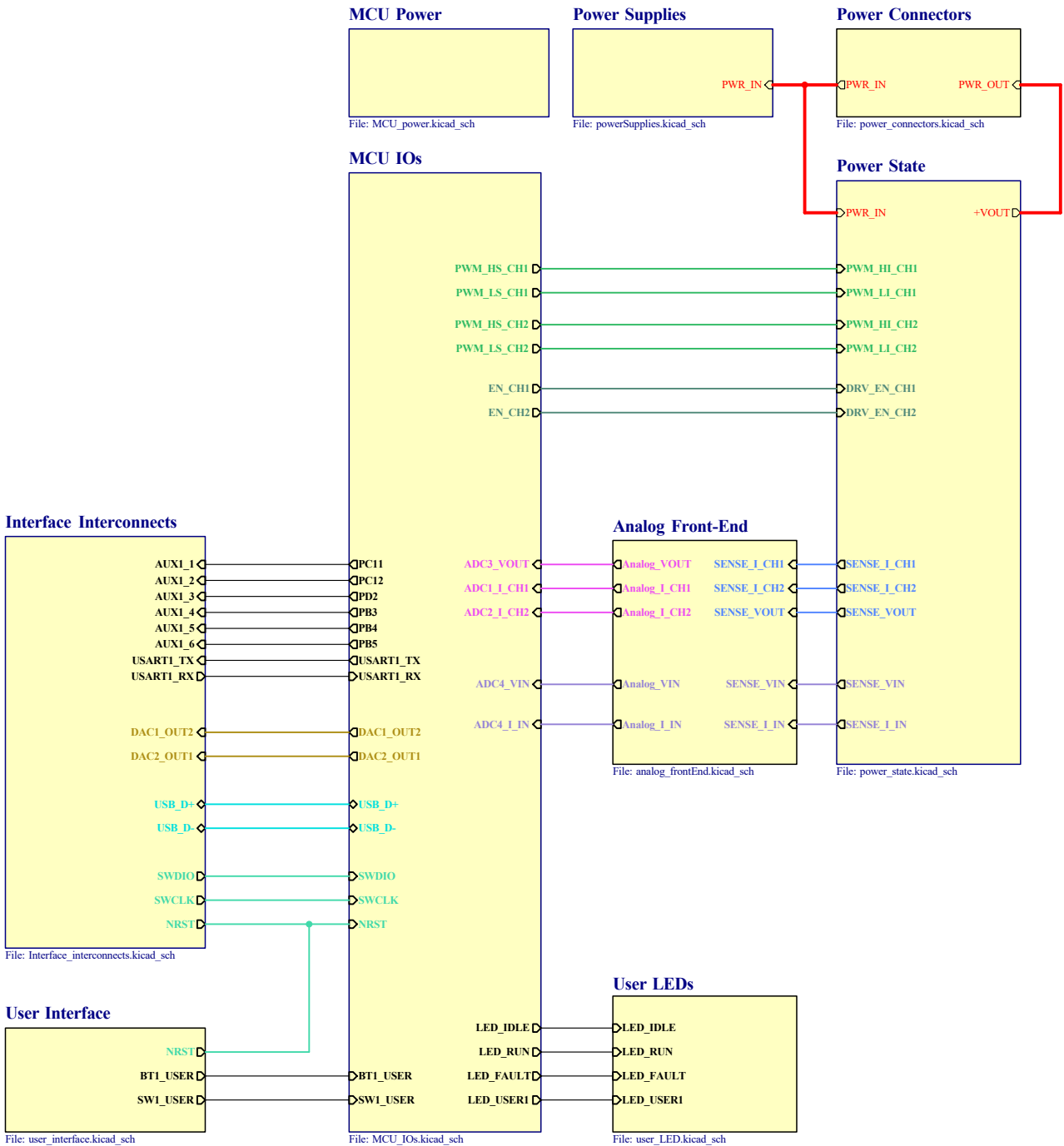
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Project Overview Cover Page	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 Date: 03/06/2026
	Sheet: /	Size: A3	Sheet: A of 9
KiCad E.D.A. 10.0.3		File: 2Phase_SynchBuck.kicad_sch	

[B] Block Diagram



Project: 2-Phase Interleaved Sync-Buck Converter			LQRx		
Title: Block Diagram (Top Level View)	Drawing by: NATCHPAI		Date: 08/04/2026	Variant:	
	Reviewed by:		Date:	Rev: 1.0	Date: 03/06/2026
	Sheet: /Block Diagram/			Size A3	Sheet:
	KiCad E.D.A. 10.0.3			File: Block_diagram.kicad_sch	B of 9
6		7		8	

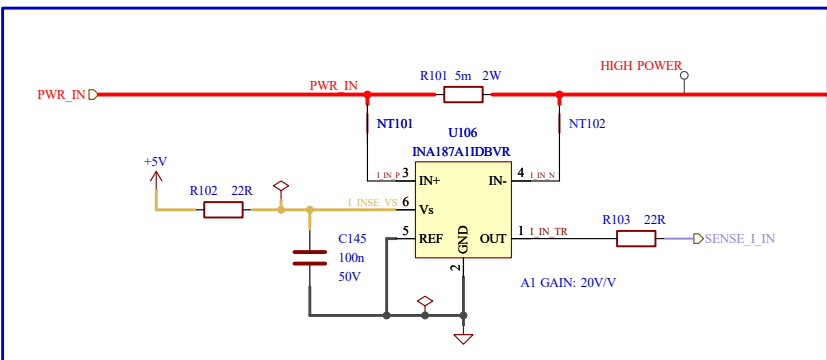
[C] Top Hierarchical View



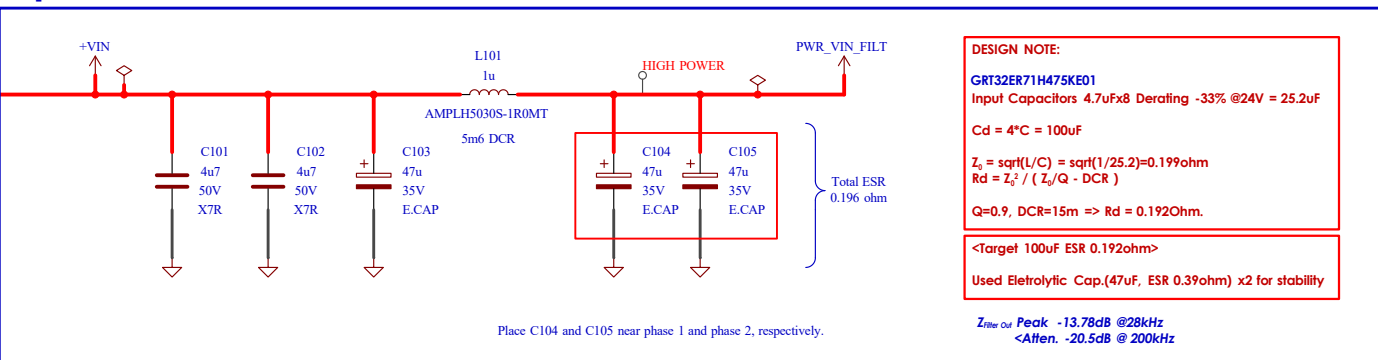
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: Top Hierarchical View (Top Level View)	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 03/06/2026
	Sheet: /Top/		Size: A3	Sheet: C of 9
KiCad E.D.A. 10.0.3		File: Top_hierarchical.kicad_sch		

[1] Power State

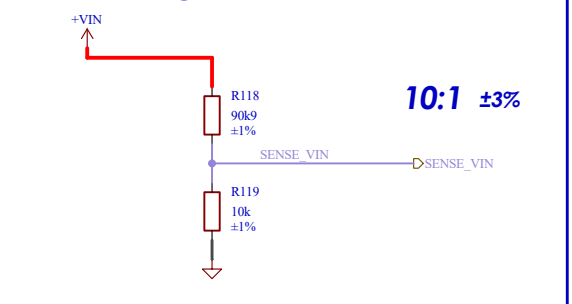
I_{IN} Sense



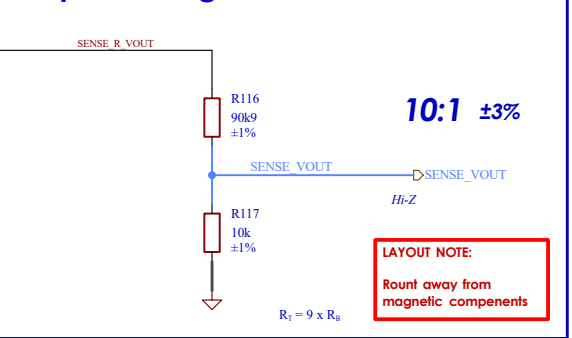
Input Filter



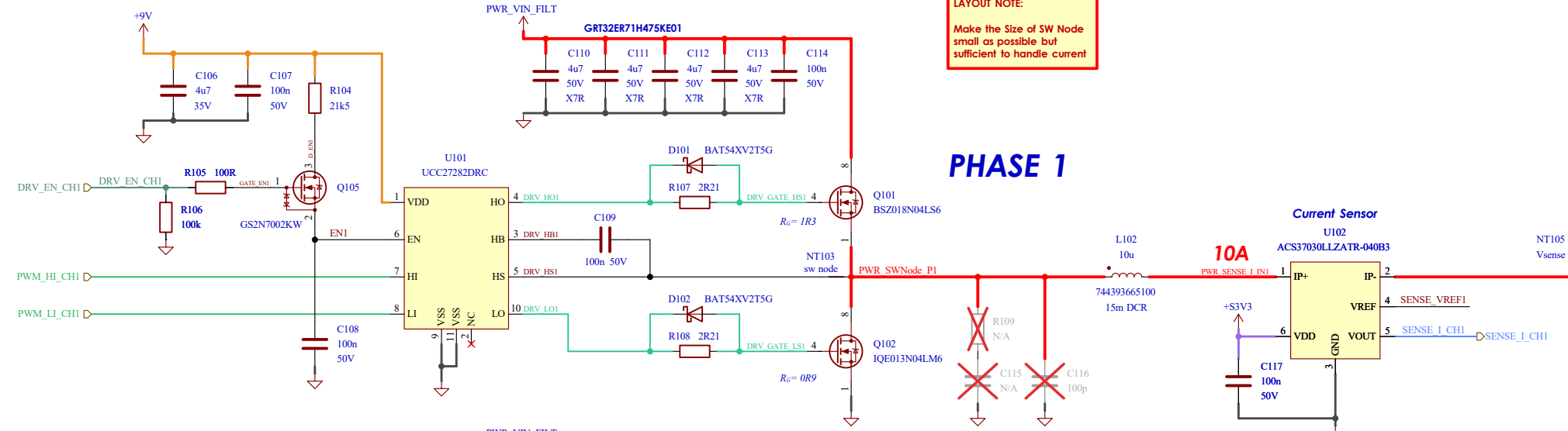
Input Voltage Sense



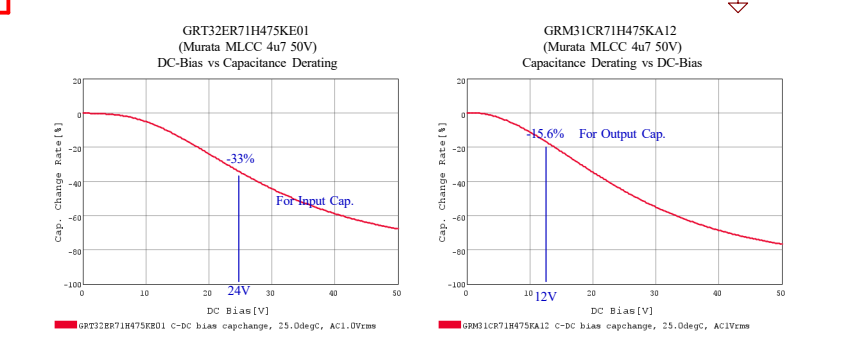
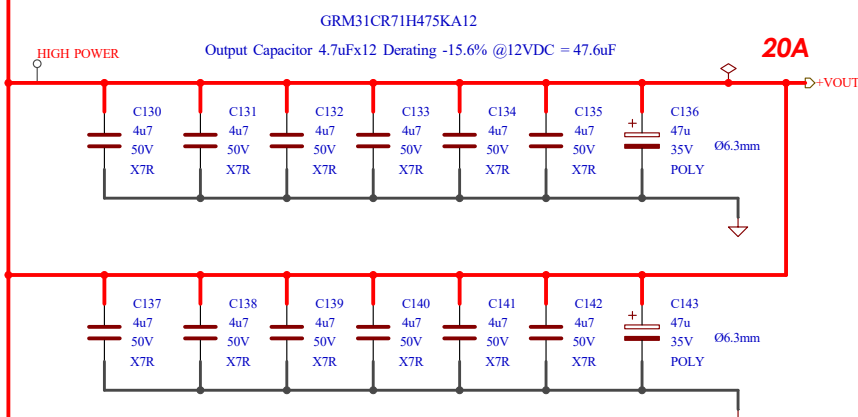
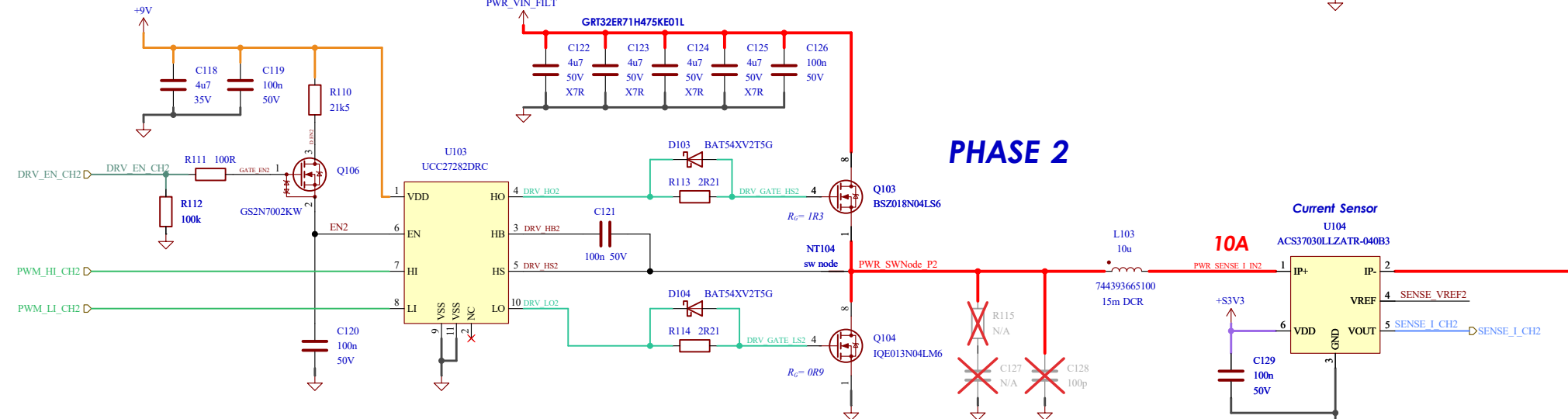
Output Voltage Sense



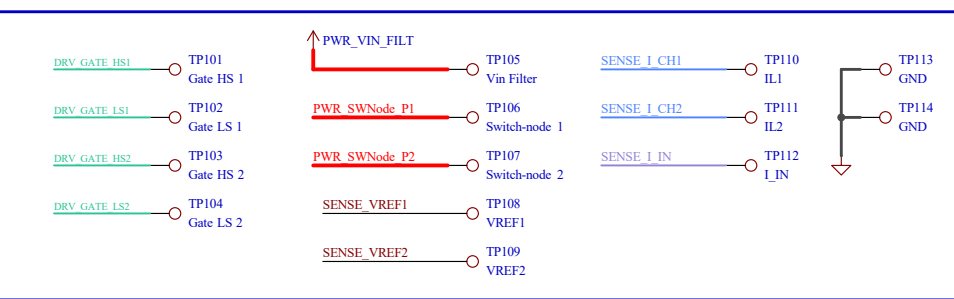
PHASE 1



PHASE 2



Test Point



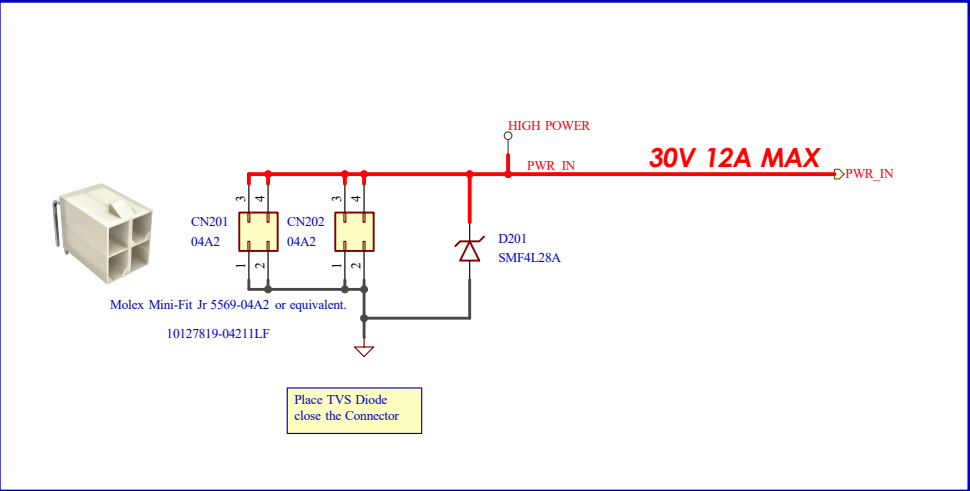
DEBUG NOTE:
If Switch-node are ringing
C116 and C128 are capacitor to find R C values in the snubber circuit.
 $C_{sn} = C_{gr} [f_2^2 / (f_1^2 - f_2^2)]$
 $L_{sn} = 1 / [2\pi f_1 C_{gr}]$
 $C_{sub} = (4 \cdot 8) C_{gr}$
 $R_{sub} = 1 / Q [\sqrt{L_{sn} / C_{sub}}]$



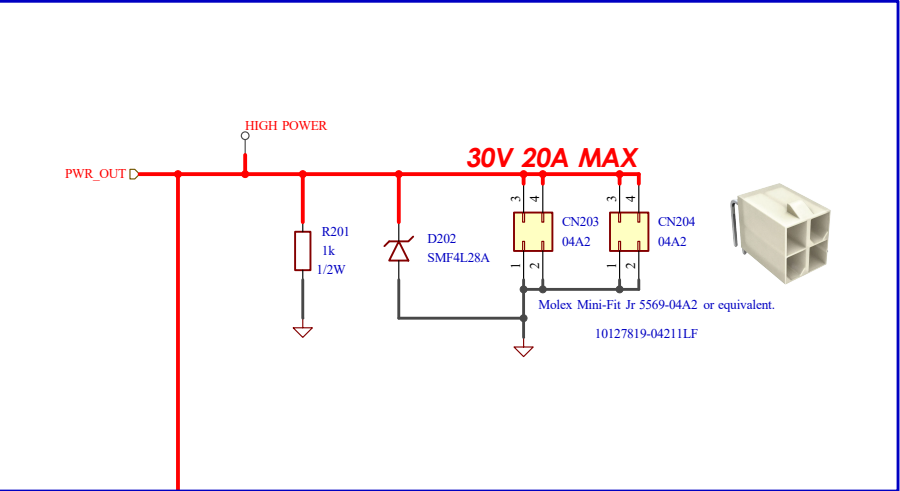
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power State		Drawing by: NATCHPAI	Date: 08/04/2026
2-Phase Buck & Gate Driver		Reviewed by:	Rev: 1.0
KiCad E.D.A. 10.0.3		Sheet: /Top/Power State/	Date: 03/06/2026
		File: power_state.kicad_sch	Size: A3
			Sheet: 1 of 9

[2] Power Connectors

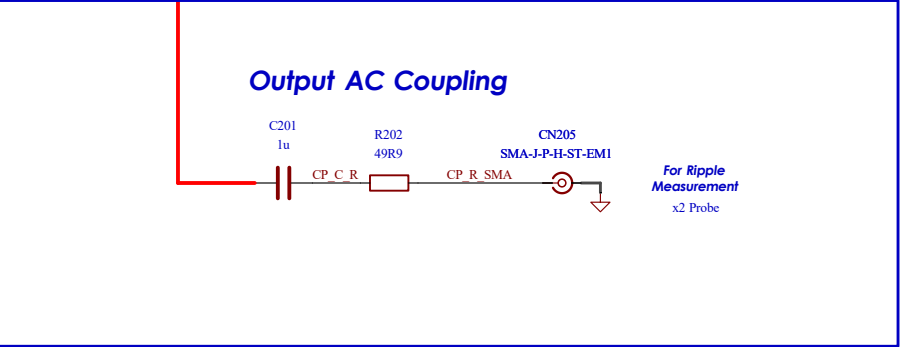
Input Connector



Output Connector



Output AC Coupling

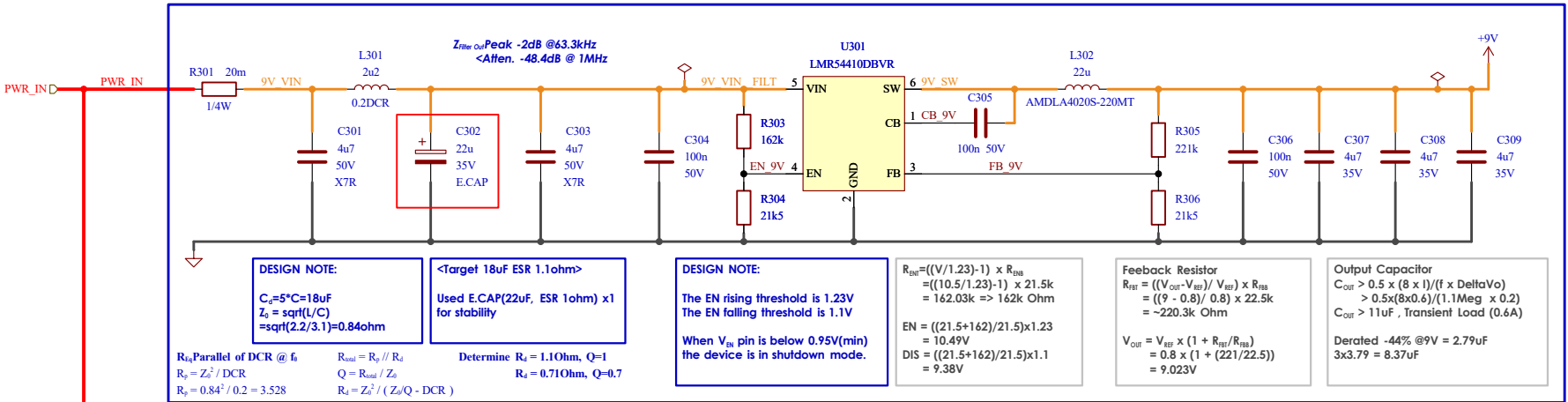


Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Connectors	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 Date: 03/06/2026
	Sheet: /Top/Power Connectors/		Size A3
KiCad E.D.A. 10.0.3	File: power_connectors.kicad_sch		Sheet: 2 of 9

[3] Power Supplies

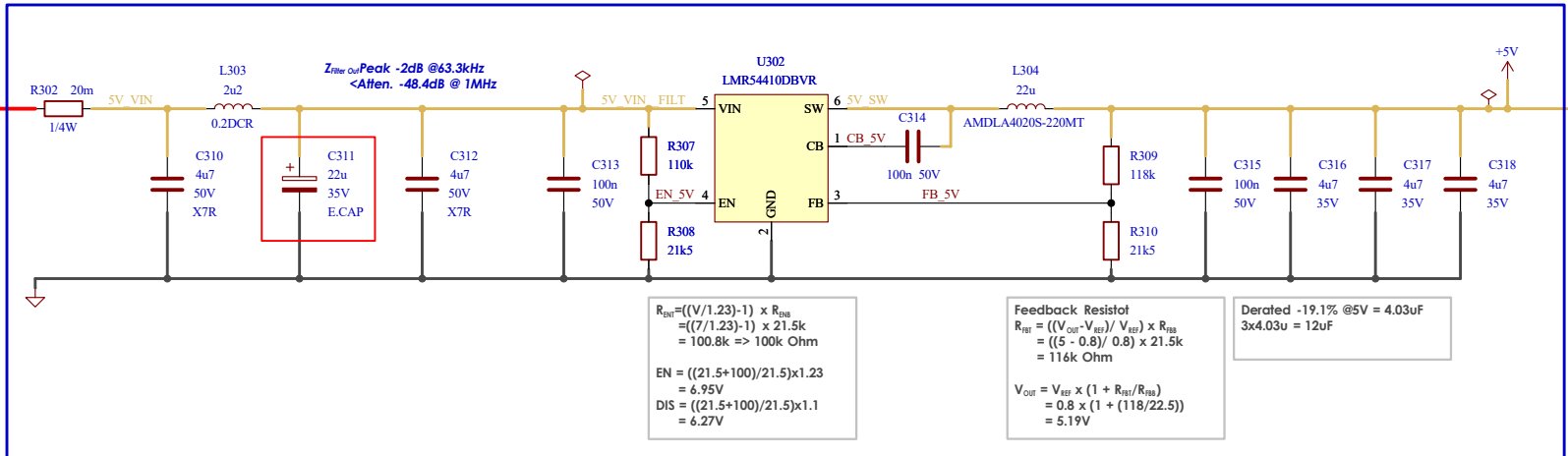
Buck Converter 9V 1A (1.1MHz)

*Enable when V_{IN} is above 11V

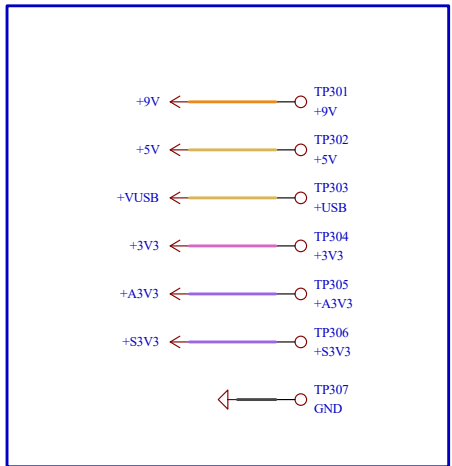


Buck Converter 5V 1A (1.1MHz)

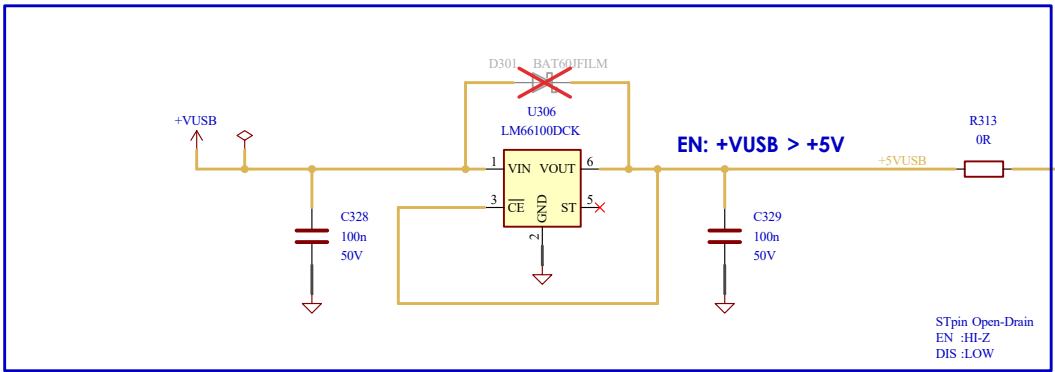
*Enable when V_{IN} is above 7V



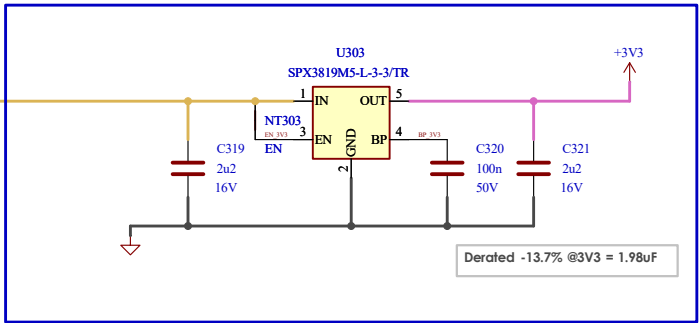
Test Point



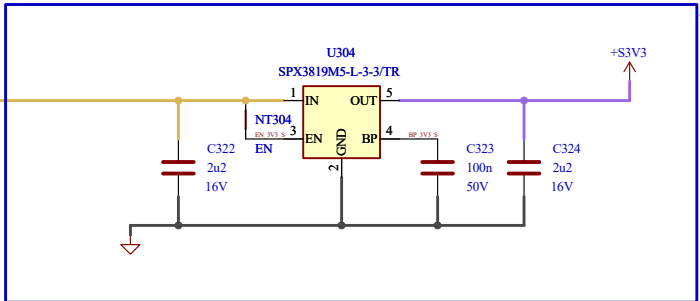
Reverse Current Blocking



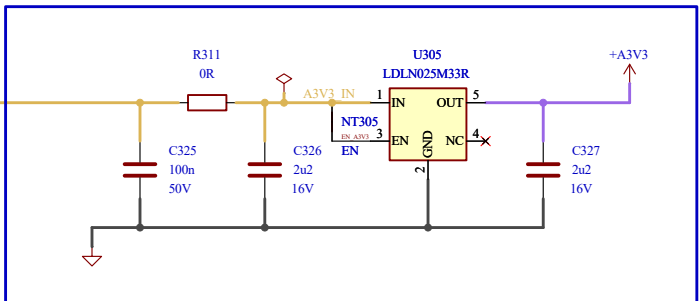
LDO 3V3 500mA (Digital path)



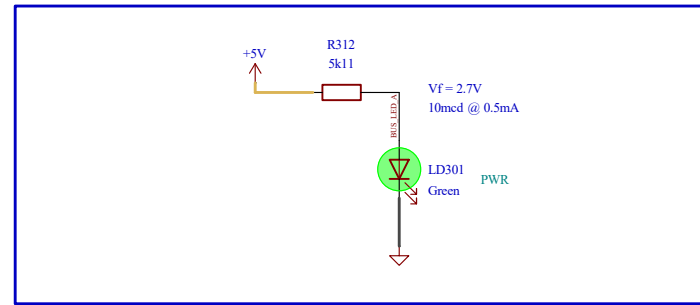
LDO 3V3 500mA (Sensor path)



LDO 3V3 250mA (Analog path)

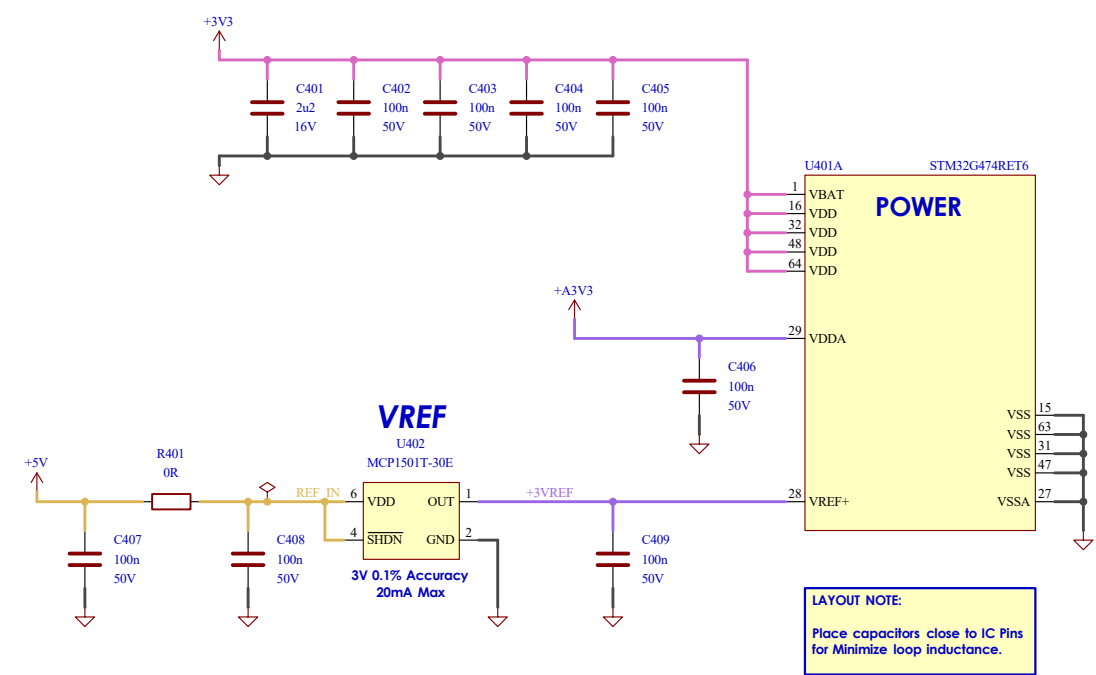


LED Indicators



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Supplies Auxiliary 9V 5V 3V3	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 Date: 03/06/2026
KiCad E.D.A. 10.0.3	Sheet: /Top/Power Supplies/ File: powerSupplies.kicad_sch		Size: A3 Sheet: 3 of 9

[4] Microcontroller Power



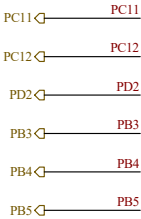
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Microcontroller Power STM32G474Rx	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 Date: 03/06/2026
	Sheet: /Top/MCU Power/		Size A3
KiCad E.D.A. 10.0.3		File: MCU_power.kicad_sch	
		4 of 9	

[5] Microcontroller IOs

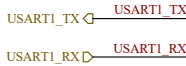
Interconnects

Auxiliary Port 1

GPIO



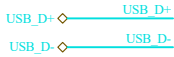
USART



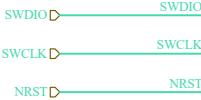
Analog Output (DAC)



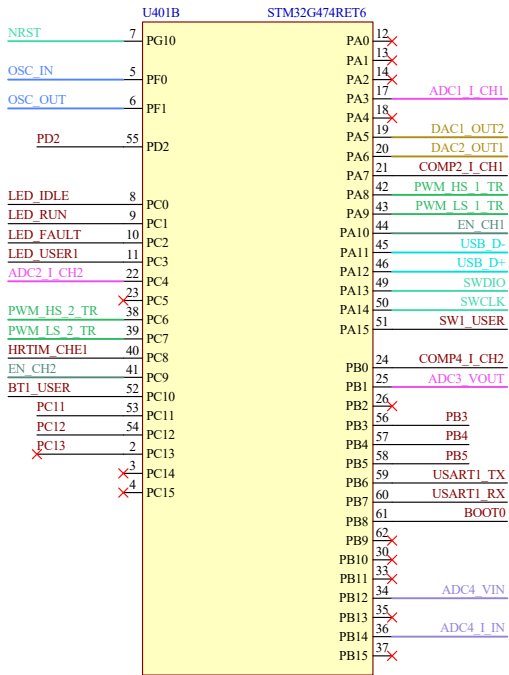
USB2.0 FS



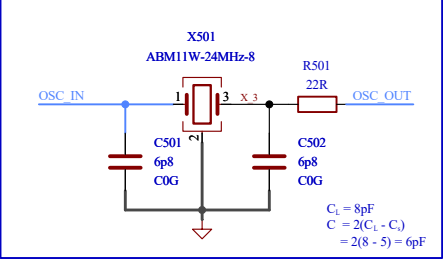
Serial Wire Debug



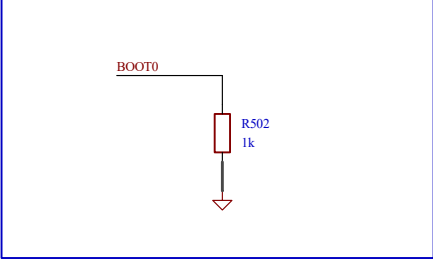
STM32G474 I/O



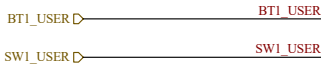
HSE Clock (24MHz)



BOOT0



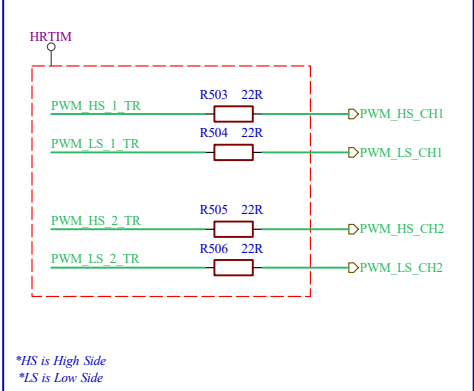
User Interface



Test Point



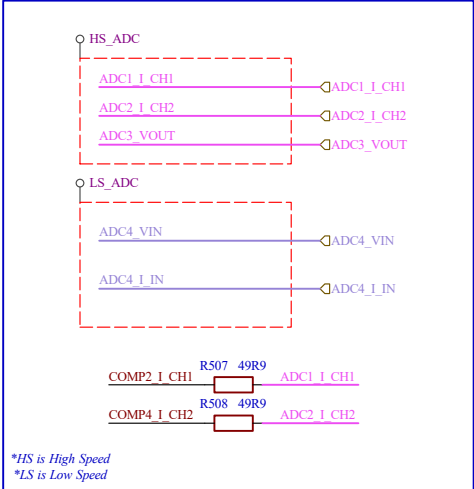
High Resolution Timer



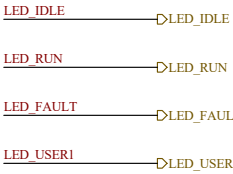
Enable Gate Driver



ADC



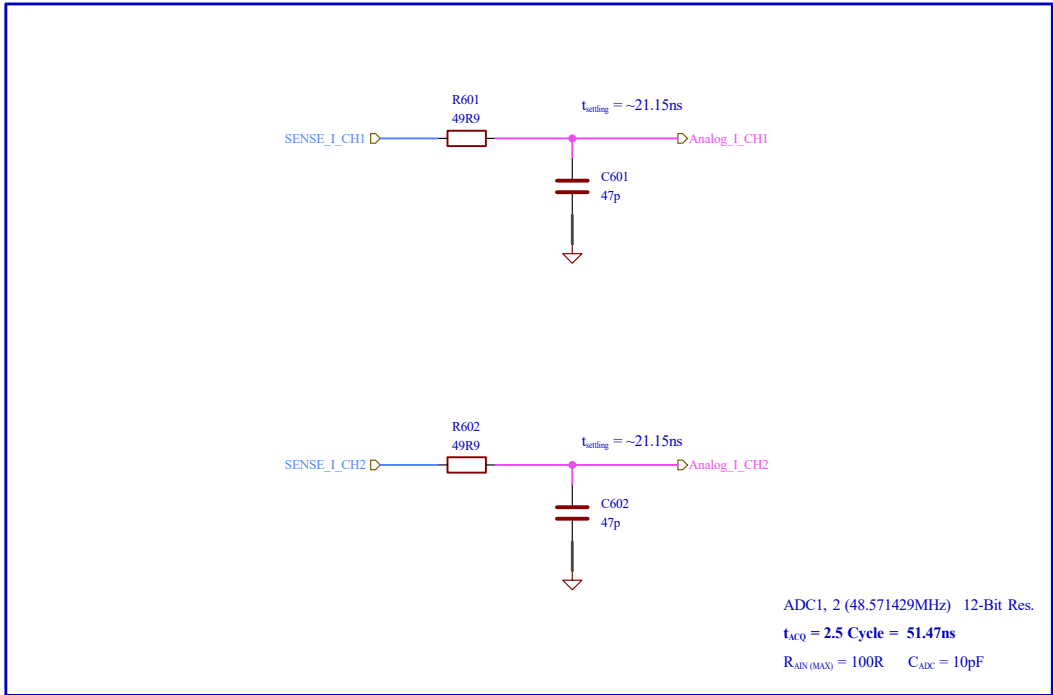
LEDs



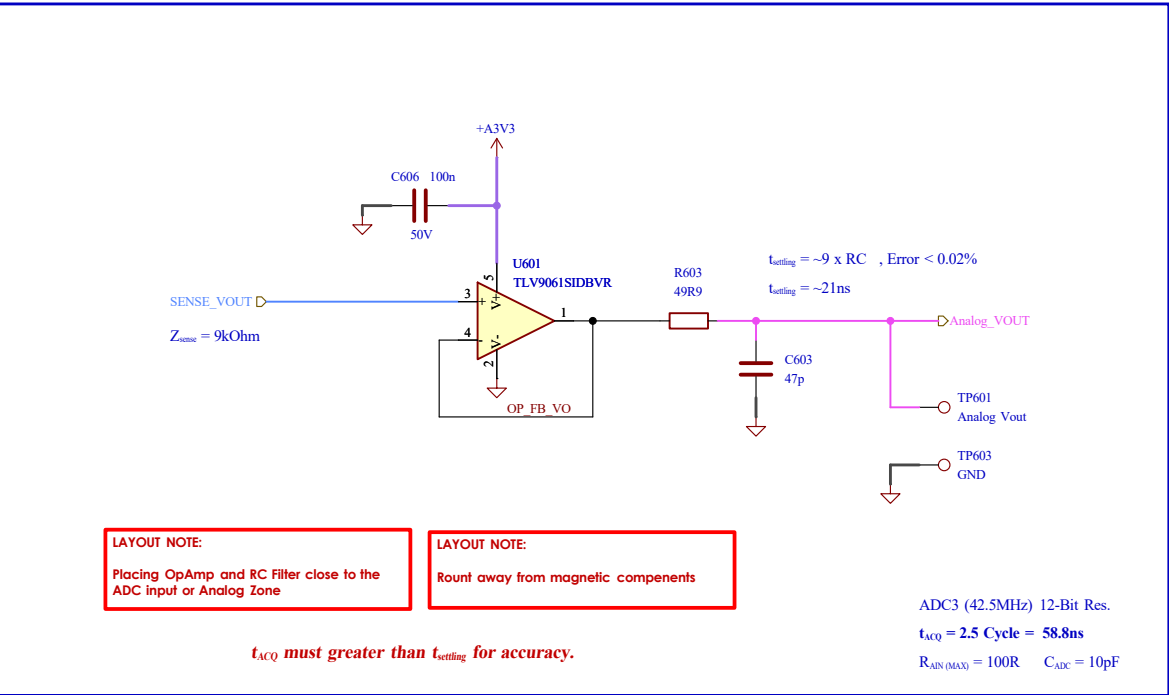
Project: 2-Phase Interleaved Sync-Buck Converter				LQRx		
Title: Microcontroller IOs STM32G474Rx	Drawing by: NATCHPAI		Date: 08/04/2026		Variant:	
	Reviewed by:		Date:		Rev: 1.0	Date: 03/06/2026
	Sheet: /Top/MCU IOs/ File: MCU_IOs.kicad_sch				Size A3	Sheet: 5 of 9
	KiCad E.D.A. 10.0.3					

[6] Analog Front-End

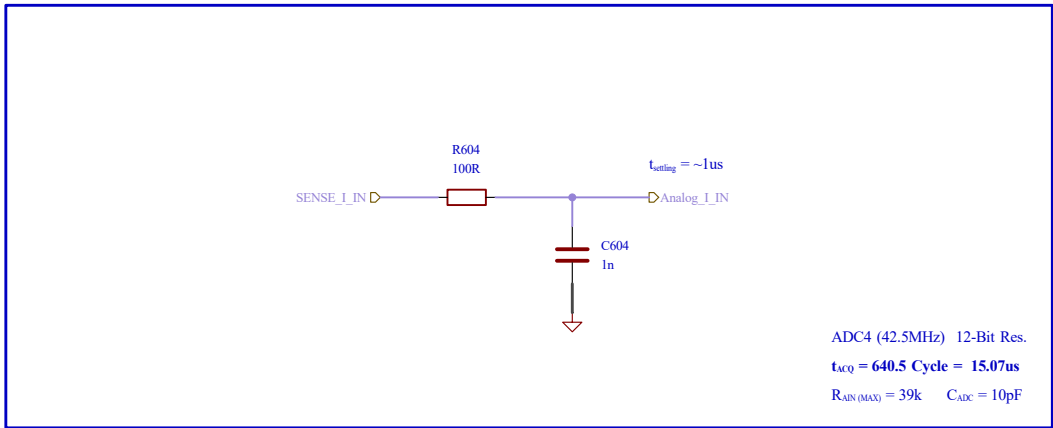
Current Path (High Speed)



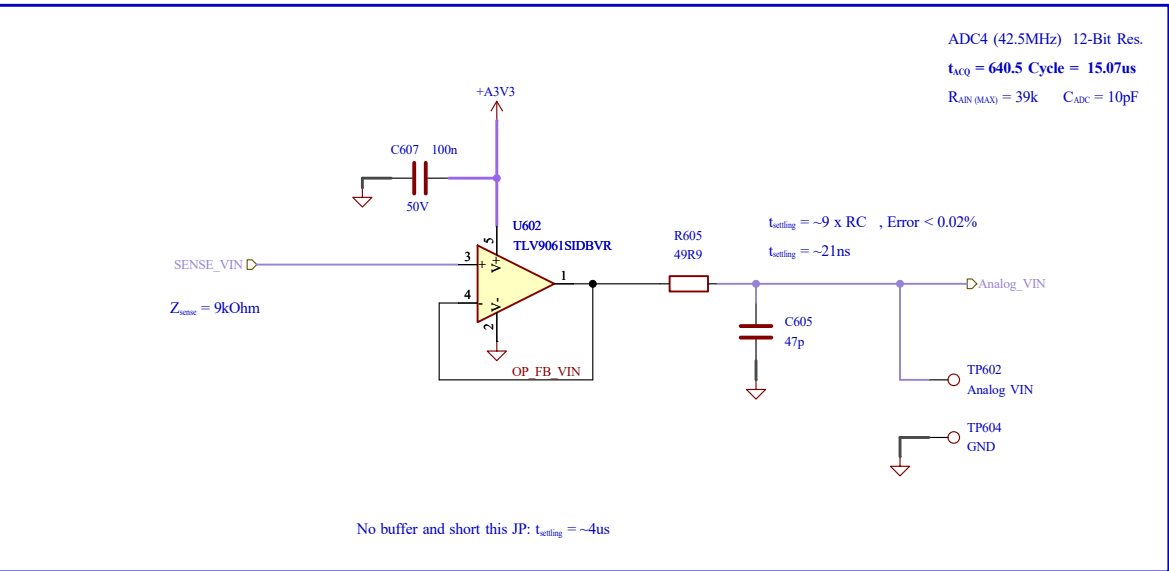
Output Voltage Path (High Speed)



Input Current Path (Low Speed)



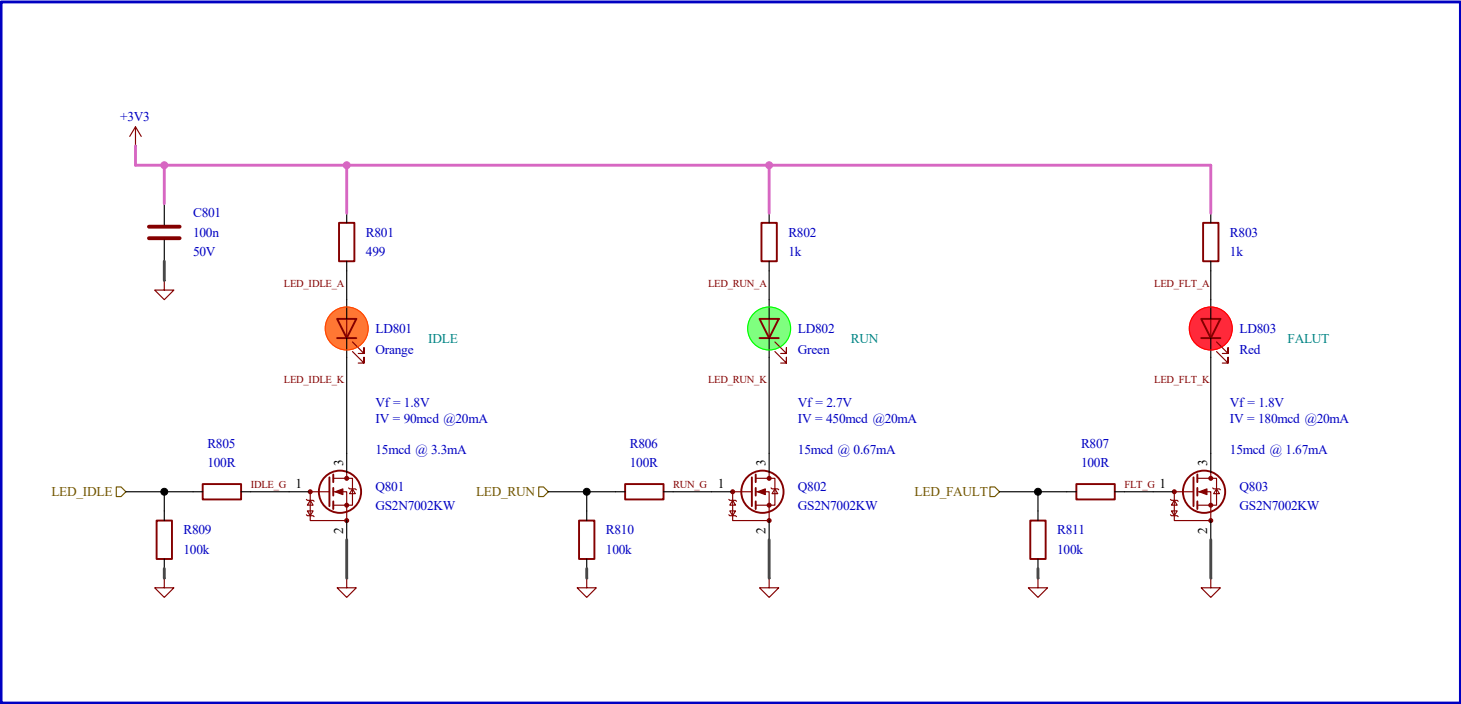
Input Voltage Path (Low Speed)



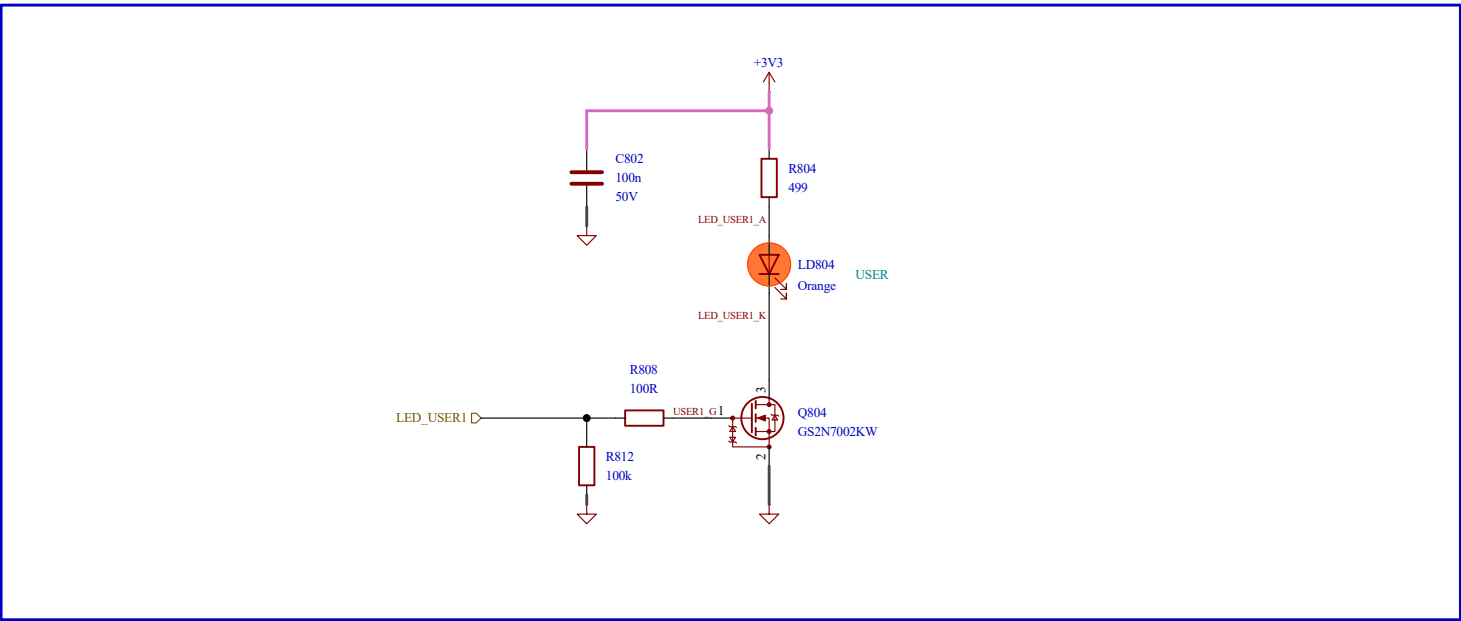
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Analog Front-End	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0
	Sheet: /Top/Analog Front-End/		Date: 03/06/2026
KiCad E.D.A. 10.0.3		File: analog_frontEnd.kicad_sch	Size: A3
			Sheet: 6 of 9

[8] User LEDs

Status LEDs



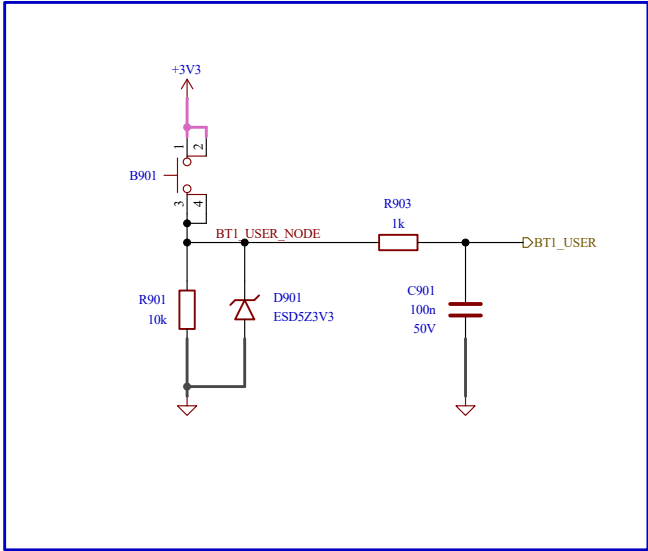
User LEDs



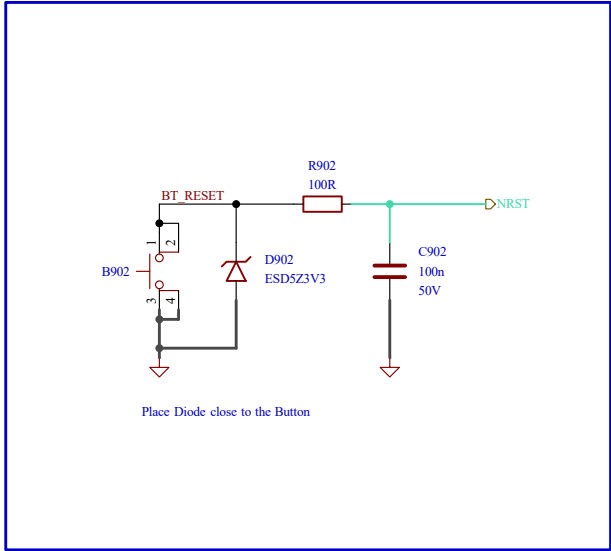
Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: User LEDs Status LEDs	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 03/06/2026
	Sheet: /Top/User LEDs/		Size A3	Sheet: 8 of 9
KiCad E.D.A. 10.0.3		File: user_LED.kicad_sch		

[9] User Interface

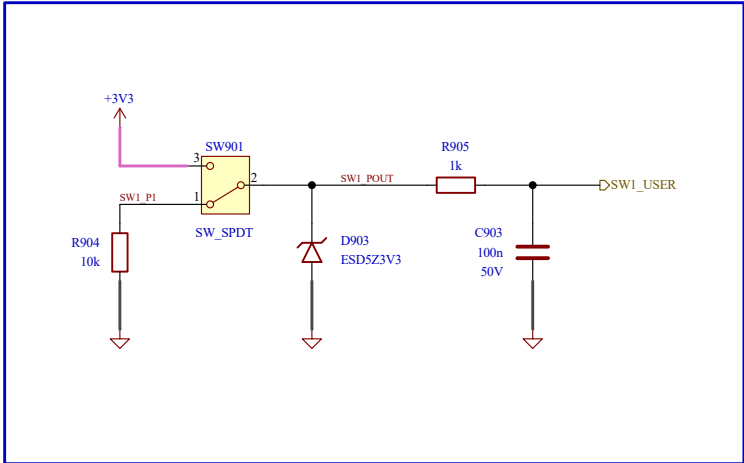
User Button



Reset Button

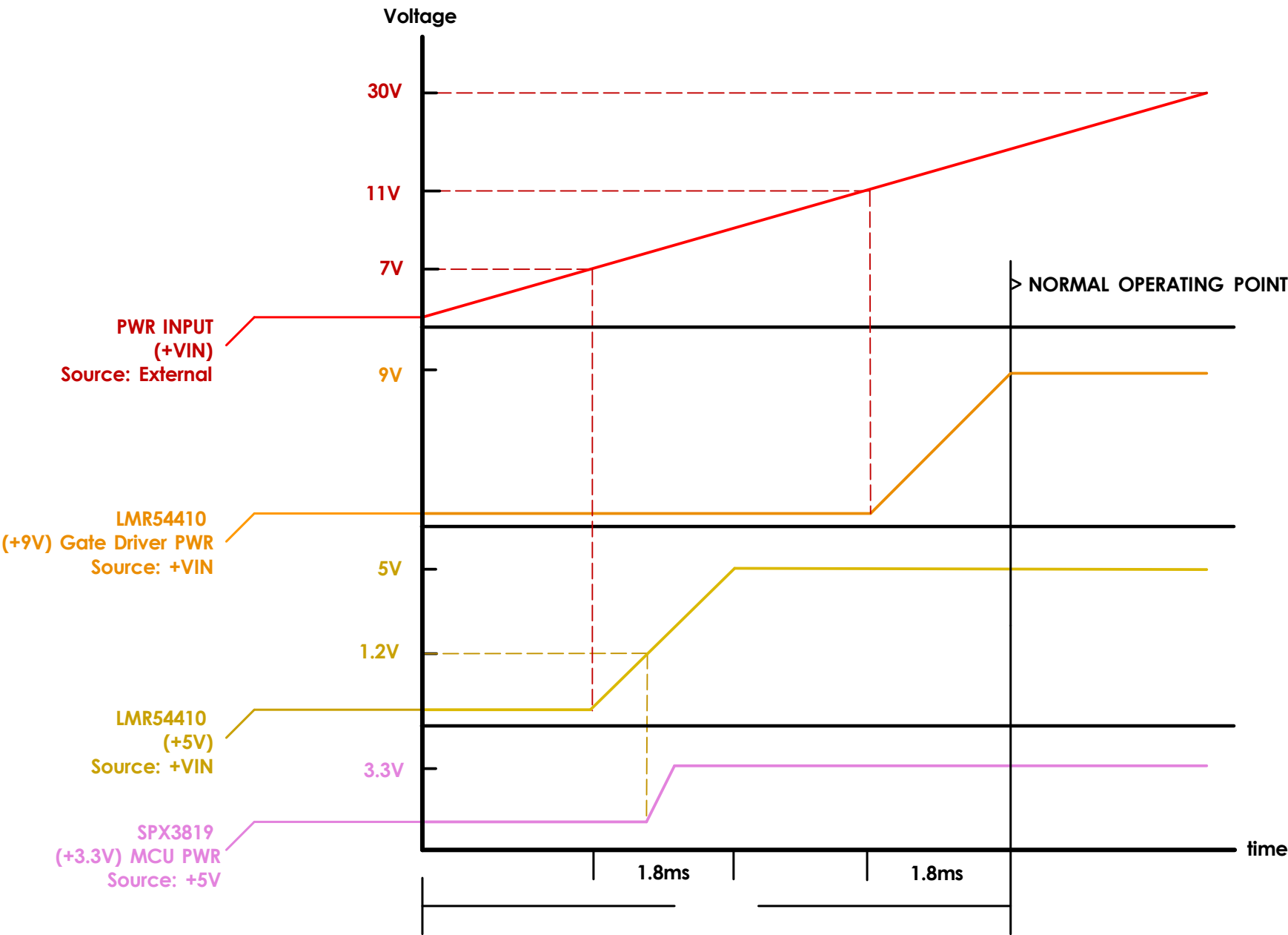


User Switch



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx		
Title: User Interface Button & Switch	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:	
	Reviewed by:	Date:	Rev: 1.0	Date: 03/06/2026
	Sheet: /Top/User Interface/		Size A3	Sheet: 9 of 9
	KiCad E.D.A. 10.0.3		File: user_interface.kicad_sch	

[D] Power Supply Sequencing



Project: 2-Phase Interleaved Sync-Buck Converter		LQRx	
Title: Power Supply Sequencing	Drawing by: NATCHPAI	Date: 08/04/2026	Variant:
	Reviewed by:	Date:	Rev: 1.0 Date: 03/06/2026
	Sheet: /Power Supply Sequencing/		Size A3 Sheet: D of 9
KiCad E.D.A. 10.0.3		File: pwr_sequencing.kicad_sch	

1	2	3	4	5	6	7	8	
[E] Revision History								
A								
B	Revision 1.0							
	- (18/05/2026) Initialize the project - (03/06/2026) PC Board is Released							
C								
D								
E								
F	1	2	3	4	5	6	7	
					Project: 2-Phase Interleaved Sync-Buck Converter			
Revision History					LQRx			
					Title:		Drawing by:	
					NATCHPAI		Date:	
					Reviewed by:		Date:	
							Rev:	
							Date:	
							1.0	
							03/06/2026	
					Sheet: /Revision History/		Size	
					File: RevHistory.kicad_sch		Sheet:	
							A3	
							E of 9	